A FILTERED COLIMIT OF SMOOTH ALGEBRAS IS A FILTERED COLIMIT OF STANDARD SMOOTH ALGEBRAS

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Editorial note. The source proof is complete; no omitted steps.

Allowed standard prerequisites (editorial).

- Colimit criterion for ind-smooth maps. Algebra, lemma when-colimit (cross-chapter, named). Reduces the claim to factoring every finitely presented A -> Lambda through a standard smooth algebra.
- Smooth algebra retracts to a standard smooth algebra. lemma-smooth-standard-smooth (carded L26799). Supplies B -> C -> B with C standard smooth over R.

Source reference. Original reference 07C3: focused-seeds/geometry/S9-smo-colimit-standard-smooth/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07CH: sources/stacks-smoothing/pinned-original-source.tex, source lines 807–807. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $R \to \Lambda$ be a ring map. If Λ is a filtered colimit of smooth R-algebras, then Λ is a filtered colimit of standard smooth R-algebras.*

Proof. Let $A \to \Lambda$ be an R-algebra map with A of finite presentation over R. According to Algebra, Lemma 07C3 we have to factor this map through a standard smooth algebra, and we know we can factor it as $A \to B \to \Lambda$ with B smooth over R. Choose an R-algebra map $B \to C$ with a retraction $C \to B$ such that C is standard smooth over R, see Lemma 07CH. Then the desired factorization is $A \to B \to C \to B \to \Lambda$. □

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